# Junjie Zhao

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## EDUCATION

**University of Rochester** **Rochester, NY**
*Bachelor of Science, Computer Science* *Anticipated May 2027*
- Current GPA: 3.99 out of 4.00

## HONORS AND AWARDS

- Dean's List: Fall 2023 to Spring 2026
- Honorable Mentions for the 2026 CRA Outstanding Undergraduate Researcher Award
- Phi Beta Kappa, Elected in Junior Year

## RESEARCH EXPERIENCE

**Department of Computer Science, University of Rochester** **Rochester, NY**
*Undergraduate Research Assistant in Horizon Lab* *Feb 2025 – Present*
*Supervised by Prof. Yuhao Zhu*
- Lead the project on an open, wearable, end-to-end wireless varifocal AR research platform for XR systems and human-centered computing research.
- Designed and prototyped the AR hardware using Fusion 360 and 3D printing, integrating a Linux-based client, IMU, outward-facing camera, eye-tracking camera, and varifocal display.
- Built a client-server sensing/rendering pipeline for wireless camera and IMU streaming, server-side localization/rendering, and frame transmission back to the wearable display.
- Conducted Zemax simulations of virtual image formation, including FOV, distortion, and third-order aberration analysis, to guide optical and system-level design trade-offs.

*Undergraduate Research Assistant in ROC-HCI Lab* *Sep 2024 – May 2026*
*Supervised by Dr. Masum Hasan and Prof. Ehsan Hoque*
- Contributed to HAL, an LLM alignment project for improving human-likeness in virtual patient dialogue and medical communication training.
- Prototyped DPO training components and supported data cleaning, synthetic dialogue generation, benchmarking, and debugging for SOPHIE-LLM.
- Built and debugged a lightweight agent framework connecting API-based and local LLM endpoints for batch synthetic data generation and structured output parsing.

**Goergen Institute for Data Science and AI, University of Rochester** **Rochester, NY**
*Undergraduate Research Assistant* *Jan 2025 – May 2025*
*Supervised by Prof. Cantay Caliskan*
- Designed an end-to-end vegetation indexing pipeline integrating Mask2Former, MiDaS, and HSV-based refinement to classify trees, plants, and grass in complex urban scenes.

## PUBLICATIONS

- Masum Hasan, Junjie Zhao, Ehsan Hoque
  HAL: Inducing Human-likeness in LLMs with Alignment
  **EMNLP 2026** *Main Conference Paper*

- Weikai Lin, Junjie Zhao, Sushant Kondguli, Carll Marshall, Yuhao Zhu
  ControlGS: Conditioning Neural Gaussians for Downstream-Processing–Aware XR Rendering
  **SIGGRAPH Asia 2026** *Conference Paper*

## FELLOWSHIPS AND GRANTS

**Schwartz Discover Grant ($5000)**
*Office of Undergraduate Research, University of Rochester* *2025*

- Project: Developing a Personality-aware Large Language Model for Human-Computer Interaction
- Mentor: Dr. Masum Hasan and Prof. Ehsan Hoque

**InnovateCS Research Grant ($5000)**
*Department of Computer Science, University of Rochester* *2026*

- Project: An Open AR Platform for Human Centered Computer System Research
- Mentor: Prof. Yuhao Zhu

## TEACHING EXPERIENCE

**Department of Computer Science, University of Rochester** **Rochester, NY**
*Open Lab Support Teaching Assistant* *Aug 2026 – Dec 2026*
*CSC 252 - Computer Organization Teaching Assistant* *Jan 2026 – May 2026*
*CSC 242 - Introduction to Artificial Intelligence Teaching Assistant* *Aug 2025 – Dec 2025*
*CSC 172 - Data Structures & Algorithms Summer Session Teaching Assistant* *June 2025 – Aug 2025*
*CSC 172 - Data Structures & Algorithms Lab Teaching Assistant* *Jan 2025 – May 2025*
*CSC 172 - Data Structures & Algorithms Workshop Leader* *Aug 2024 – Dec 2024*

## INTERSHIP EXPERIENCE

**Institute of Automation, Chinese Academy of Sciences** **Beijing, China**
*Artificial Intelligence Intern* *May 2024 – Aug 2024*

- Deployed the YOLOv5 object detection model on NVIDIA Jetson Nano board; Optimized performance for edge-based inference.
- Integrated the model with a monocular camera to accurately convey various road traffic signs, including directional arrows, traffic light, double amber lines, and so on.
- Designed and integrated control algorithms with the detection model to enable a smart car to automatically execute actions such as braking, steering, turning, and correcting its direction in real time.

## RESEARCH PRESENTATIONS

**NY Architecture & Systems Day, Princeton University** **Princeton, NJ**
*Lightning Talk & Poster Presenter* *May 2026*

- Presented "End-to-End Wireless Varifocal AR Prototype," demonstrating world-locked rendering and multi-stream sensing on a wearable AR research platform.

## CAMPUS ACTIVITIES

**Department of Computer Science, University of Rochester** **Rochester, NY**
*Computer Science Undergraduate Council Tutor* *Aug 2024 – May 2025*

- Provided one-to-one and small-group tutoring in algorithms, data structures, and programming problem-solving.

## SKILLS

- Programming Languages: Java, Python, C, HTML, JavaScript, Assembly
- Softwares: Photoshop, Premiere Pro, Autodesk Fusion, Zemax
- Machine Learning and AI Tools: Pytorch, Neural Network Model Development, Fine-Tuning, DPO, RLHF, SFT